

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – I
PAPER –1
TEST DATE: 23-12-2021

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.

Section – A (01 – 04, 20 – 23, 39 – 42): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 –10, 24 – 29, 43 – 48): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. In an X-ray set up accelerating potential difference is set at V volt and first a target metal having atomic no. Z_1 is used then target metal having Z_2 is used (λ_{α_1} is wavelength of K_{α} X-ray from Z_1 target and λ_{α_2} is wavelength of K_{α} x-ray with Z_2 target)

- (A) If $Z_1 > Z_2$ then $(\lambda_{\alpha_1} - \lambda_{\min}) > (\lambda_{\alpha_2} - \lambda_{\min})$
- (B) If $Z_1 < Z_2$ then $(\lambda_{\alpha_1} - \lambda_{\min}) > (\lambda_{\alpha_2} - \lambda_{\min})$
- (C) The difference $(\lambda_{\alpha_1} - \lambda_{\min})$ and $(\lambda_{\alpha_2} - \lambda_{\min})$ will be same for both the target metals.
- (D) If $Z_1 \leq Z_2$ then $(\lambda_{\alpha_1} - \lambda_{\min}) \leq (\lambda_{\alpha_2} - \lambda_{\min})$

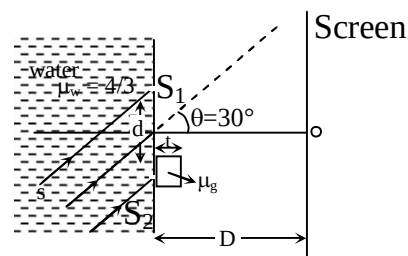
Ans. B

Sol.
$$\frac{1}{\lambda} = R(Z - b)^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

For K_{α} transition
$$\frac{1}{\lambda_{\alpha}} = R(Z - 1)^2 \left[1 - \frac{1}{4} \right]$$

$$\lambda_{\alpha} - \lambda_{\min} \text{ depends on } \frac{1}{3R(Z - 1)^2}$$

2. In a YDSE experiment, the two slits are covered with a transparent membrane of negligible thickness which allows light to pass through it but does not allow water. A glass slab of thickness $t = 0.41$ mm and refractive index $\mu_g = 1.5$ is placed in front of one of the slits as shown in the figure. The separation between the slits is $d = 0.30$ mm. The entire space to the left of the slits is filled with water of refractive index $\mu_w = 4/3$.



A coherent light of intensity I and absolute wavelength $\lambda = 5000 \text{ \AA}$ is being incident on the slits making an angle 30° with horizontal. If screen is placed at a distance $D = 1\text{m}$ from the slits, then the position of central maxima will be

- (A) 0.58 cm above the central line of slits
- (B) 0.58 cm below the central line of slits
- (C) 1.66 cm above the central line of slits
- (D) 1.66 cm below the central line of slits

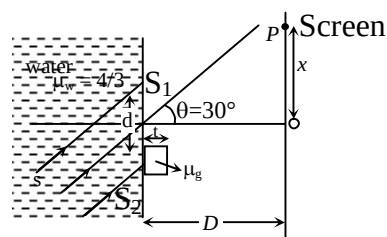
Ans. D

Sol. If central maxima is formed at point P on the screen, then

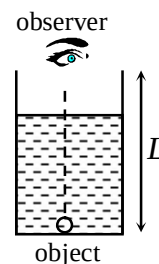
$$\text{Path difference} = S_2P + (\mu_g - 1)t - S_1P - \mu_w d \sin 30^\circ = 0$$

$$\Rightarrow S_2P - S_1P = \frac{xd}{D} = \mu_w d \sin 30^\circ - (\mu_g - 1) \times t$$

$$\Rightarrow x = -1.66 \text{ cm}$$



3. A cylinder is filled with a liquid of refractive index μ . The radius of the cylinder is decreasing at a constant rate K . The volume of the liquid inside the container remains constant at V . The observer and the object O are in a state of rest and at a distance L from each other. The apparent velocity of the object as seen by the observer, (when radius of cylinder is r)



(A) $\frac{(1-\mu)2KV}{(\pi\mu r^3)}$

(B) $\frac{(1-\mu)2KV}{(\pi\mu Lr^2)}$

(C) $\frac{(1-\mu)2K}{\mu}$

(D) $\frac{(1-\mu)K}{2\mu}$

Ans. A

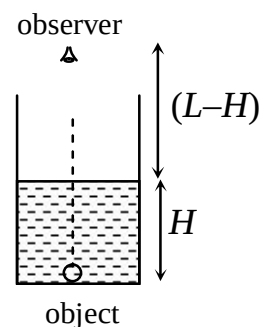
Sol. $X_{app} = L - H + \frac{H}{\mu}$

$$\frac{dX_{app}}{dt} = \frac{dH}{dt} \left(\frac{1-\mu}{\mu} \right)$$

$$\pi r^2 H = V$$

$$\therefore \frac{dH}{dt} = -\frac{2H}{r} \frac{dr}{dt} = \frac{2KH}{r}$$

$$\therefore \frac{dX_{app}}{dt} = \frac{2KV}{\pi r^3} \left(\frac{1-\mu}{\mu} \right)$$



4. A car starts from rest from origin on a straight line with an acceleration (a) given by the relation $a = \frac{25}{(x+2)^3}$, where a is in m/s^2 and x is in metre. The maximum velocity of the car will be (x is the position of the car)

- (A) 2.5 m/s
(B) 5 m/s
(C) 10 m/s
(D) Infinite

Ans. A

Sol. $a = v \frac{dv}{dx} = \frac{25}{(x+2)^3}$, $\frac{v^2}{2} = 25 \times \left[-\frac{1}{2(x+2)^2} \right]_0^x$, $v^2 = 25 \left[\frac{1}{4} - \frac{1}{(x+2)^2} \right]$
 $v = \sqrt{25 \left[\frac{1}{4} - \frac{1}{(x+2)^2} \right]}$, $v_{\max} = \frac{5}{2} = 2.5 \text{ m/s (at } x = \infty)$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

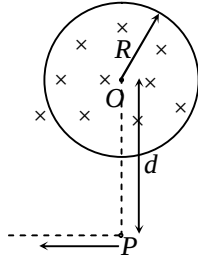
5. An electron in H-atom jumps from second excited state to first excited state and then from first excited to ground state. Let the ratio of wavelength, momentum and energy of photons emitted in these two cases be a , b and c respectively. Then

- (A) $c = \frac{1}{a}$
(B) $a = \frac{9}{4}$
(C) $b = \frac{5}{27}$
(D) $c = \frac{5}{27}$

Ans. A, C, D

Sol. As $\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ and $P = \frac{h}{\lambda}$
 Also, $E = \frac{hc}{\lambda}$

6. In a cylindrical region of radius R , there exists a time varying magnetic field B such that $\frac{dB}{dt} = k (> 0)$. A charged particle having charge q is placed at the point P at a distance $d (> R)$ from its centre O . Now, the particle is moved in the direction perpendicular to OP (see figure) by an external agent upto infinity so that there is no gain in kinetic energy of the charged particle. Choose the correct statement/s.



- (A) Work done by external agent is $\frac{q\pi R^2}{4}k$ if $d = 2R$
- (B) Work done by external agent is $\frac{q\pi R^2}{8}k$ if $d = 4R$
- (C) Work done by external agent is $\frac{q\pi R^2}{4}k$ if $d = 4R$
- (D) Work done by external agent is $\frac{q\pi R^2}{4}k$ if $d = 6R$

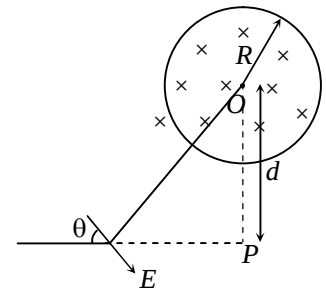
Ans. A, C, D

Sol.:
$$\int \vec{E} \cdot d\vec{l} = A \frac{dB}{dt}$$

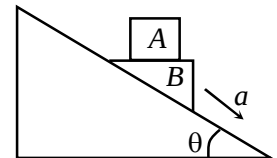
$$E 2\pi \sqrt{x^2 + d^2} = \pi R^2 k$$

$$E = \frac{\pi R^2 k}{2\sqrt{x^2 + d^2}}$$

$$W_{\text{ext}} = \int_0^\infty q\vec{E} \cdot d\vec{x} = \frac{q\pi R^2}{4}k$$



7. In the given figure a block A rests on a smooth triangular block B and the block B is given an acceleration of $a = 2 \text{ m/s}^2$ along the plane



- (A) normal force on block A due to block B is $m(g - a)$
- (B) acceleration of block A relative to block B is $a \cos \theta$.

- (C) If friction is present between block and A and B, the coefficient of friction should be greater than $\frac{a}{g} \cos \theta$, for no relative motion between A and B
- (D) If friction is present between block and A and B, the coefficient of friction should be greater than $\frac{a \cos \theta}{g - a \sin \theta}$, for no relative motion between A and B.

Ans. B, D

Sol. From FBD of A with respect to B

$$a_v = 0$$

$$mg = N - ma \sin \theta$$

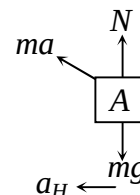
$$\Rightarrow N = mg - ma \sin \theta$$

$$ma \cos \theta = ma_H \Rightarrow a_H = a \cos \theta$$

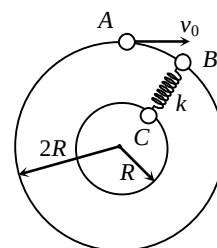
If block B is having friction then, for $a_H = 0$

$$ma \cos \theta \leq \mu N = \mu (mg - ma \sin \theta)$$

$$\mu \geq \frac{a \cos \theta}{g - a \sin \theta}$$



8. Three particles each of mass m , can slide on fixed frictionless circular tracks in the same horizontal plane as shown. Particle A moves with velocity v_0 and hits particle B elastically. Assuming that B and C are initially at rest and lie along a radial line and the spring is initially relaxed before impact, then



- (A) the velocity of B immediately after impact is v_0
- (B) the velocity of C when the stretch in the spring is maximum is $\frac{2v_0}{5}$
- (C) the velocity of B when the stretch in the spring is maximum is $\frac{4v_0}{5}$
- (D) the maximum stretch in the spring in the spring is $\sqrt{\frac{m}{5k}} v_0$

Ans.

Sol. At the time of maximum elongation angular speed of B and C are equal, let speed of B is $2v$ and C is v , By conserving angular momentum of the system about the centre

$$mv_0 2R = m2v(2R) + mv(R)$$

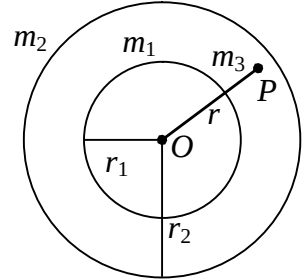
$$v = \frac{2v_0}{5}, \quad v_B = \frac{4v_0}{5}, \quad v_C = \frac{2v_0}{5}$$

Conserving energy of the system $\frac{1}{2}mv_0^2 = \frac{1}{2}kx_{\max}^2 + \frac{1}{2}m\left(\frac{4v_0}{5}\right)^2 + \frac{1}{2}m\left(\frac{2v_0}{5}\right)^2$

$$\therefore x_{\max} = \sqrt{\frac{m}{5k}}v_0$$

$\therefore$ (a), (b), (c) and (d)

9. Two concentric spherical shells masses m_1 and m_2 and radii r_1 and r_2 . Then

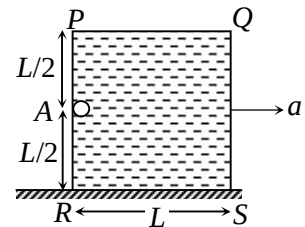


- (A) outer shell will have no contribution in gravitational field at point P
- (B) force on P is directed towards O
- (C) force on P is $\frac{Gm_1m_2}{r^2}$
- (D) force on P is $\frac{Gm_1m_3}{r^2}$

Ans. A, B, D

Sol. Gravitational field intensity inside the spherical shell is zero. Gravitational force is always directed towards the source mass. Force on P act only due to m_1 because point P is outside the spherical shell of mass m_1 . So choice (a), (b) and (d) are correct.

10. A small solid ball of density ρ is held inside at point A a cubical container of side L , filled with an ideal liquid of density 4ρ as shown in the figure. Now, if the container starts moving with constant acceleration a horizontally and the ball is released from point A simultaneously, then



- (A) For ball to hit the top of container at end Q , $a = 3g$
- (B) For ball to hit the top of container at end Q , $a = 2g$
- (C) Ball hits the top of container at end Q after a time $t = \sqrt{\frac{L}{3g}}$
- (D) Ball hits the top of container at end Q after a time $t = \sqrt{\frac{2L}{3g}}$

Ans. B, C

Sol. F.B.D. of ball in frame of container

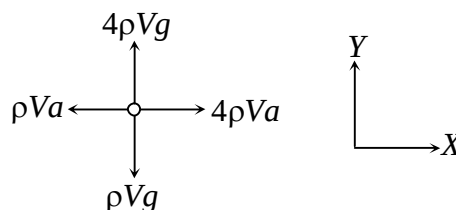
$$a_y = \frac{4\rho Vg - \rho Vg}{\rho V} = 3g$$

$$\frac{L}{2} = \frac{1}{2} 3gt^2 \Rightarrow t = \sqrt{\frac{L}{3g}}$$

$$a_x = \frac{4\rho Va - \rho Va}{\rho V} = 3a$$

$$L = \frac{1}{2} 3at^2 \Rightarrow t = \sqrt{\frac{2L}{3a}}$$

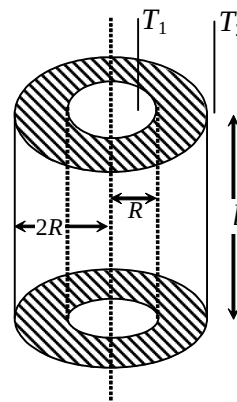
$$\text{Ball to collide at point Q, } t = \sqrt{\frac{2L}{3a}} = \sqrt{\frac{L}{3g}} \Rightarrow a = 2g$$



Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. Inner surface of a cylindrical shell of length l and of material of thermal conductivity k is kept at constant temperature T_1 and outer surface of the cylinder is kept at constant temperature T_2 such that $(T_1 > T_2)$ as shown in figure. Heat flows from inner surface to outer surface radially outward. Inner and outer radii of the shell are R and $2R$ respectively. Due to lack of space this cylinder has to be replaced by a smaller cylinder of length $\frac{l}{2}$, inner and outer radii $\frac{R}{4}$ and R respectively and thermal conductivity of material nk . If rate of radial outward heat flow remains same for same temperatures of inner and outer surface i.e. T_1 and T_2 , then find the value of n .

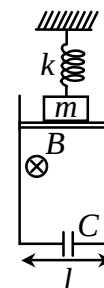


Ans. 4

Sol. Since rate of heat flow remains same in both the cases, so

$$\int_R^{2R} \frac{dx}{k 2\pi x l} = \int_{\frac{R}{4}}^R \frac{dx}{nk(2\pi x) \frac{l}{2}} \Rightarrow nk = 4k \Rightarrow n = 4$$

12. A block is attached to the ceiling by a spring that has a force constant $k = 200$ N/m. A conducting rod is rigidly attached to the block. The combined mass of the block and the rod is $m = 0.3$ kg. The rod can slide without friction along two vertical parallel rails, which are a distance $l = 1$ m apart. A capacitor of known capacitance $C = 500 \mu\text{F}$ is attached to the rails by the wires. The entire system is placed in a uniform magnetic field $B = 20$ T directed as shown. Find the angular frequency (in rad/sec) of the vertical oscillations of the block. Neglect the self-inductance and electrical resistance of the rod and all wires.



Ans. 20

Sol. Total energy $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 + \frac{1}{2}\frac{q^2}{C}$, $e = Bvl = \frac{q}{C}$

$$E = \frac{1}{2}(m + B^2l^2C)v^2 + \frac{1}{2}kx^2 \text{ as } E = \text{constant} \quad \frac{dE}{dt} = 0, \quad \frac{dv}{dt} = -\left(\frac{k}{m + B^2l^2C}\right)x,$$

$$\omega = \sqrt{\frac{k}{m + B^2l^2C}} = 20 \text{ rad/sec.}$$

13. Find the effective value or rms value of an alternating current that changes according to the law (All quantities are in S.I. unit and symbols have their usual meaning)

$$I = 10, \text{ when } 0 < t < T/8; \quad I = 0, \text{ when } \frac{T}{8} < t < \frac{T}{2}$$

$$I = -10, \text{ when } \frac{T}{2} < t < \frac{5}{8}T; \quad I = 0, \text{ when } \frac{5}{8}T < t < T; \quad I = 10, \text{ when } T < t < \frac{9}{8}T$$

Ans. 5

Sol. We solve this problem using the fact that the effective value of an alternating current is the value of a direct current that produces the same quantity of heat in a conductor the alternating current during the same time

The heat produced by the AC current during one time period from 0 to T is

$$H = 10^2 R \frac{T}{8} + 0 + 10^2 R \frac{T}{8} + 0$$

$$H = 10^2 R \frac{T}{4} \quad \dots(1)$$

In the same time interval 0 to T heat produced by constant direct current I , would have been

$$H' = I^2 RT \quad \dots(2)$$

$$H = H' \Rightarrow I^2 RT = 10^2 R \frac{T}{4}$$

$$\Rightarrow I = \frac{10}{2}$$

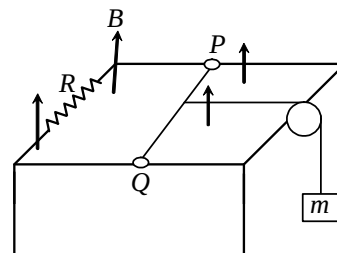
effective value or rms value of required current is 5

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15
Question Stem

A U-shape wire is formed by connecting a resistance R with two parallel conducting rod at its both ends and kept on a horizontal table. A sliding wire PQ of length l and mass m can slide over the U-Shape wire without friction. This rod PQ is connected to a block of mass m through an insulating string going over a smooth pulley as shown. A vertical magnetic field B is switched on and rod PQ is allowed to move from rest. After some time it attains a constant velocity v_0 (also called terminal velocity). Now mass of the block is tripled, it attain another terminal velocity v_T .



(Given $B = \sqrt{10}T$, $l = 1\text{ m}$, $v_0 = 10\text{ m/s}$, $R = 10\Omega$, $g = 10\text{ m/s}^2$)

14. The value of mass m is

Ans. 1.00

15. The value of terminal velocity v_T is

Ans. 30.00

Sol. (Q.14 & 15)

$$\text{Since velocity is constant, } F_{\text{net}} = 0 \Rightarrow mg = ilB = \frac{B^2 l^2 v_0}{R} \quad \left[\because i = \frac{\varepsilon}{R} = \frac{Blv_0}{R} \right]$$

$$\text{On solving we get, } m = 1\text{ kg}$$

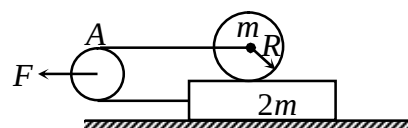
$$\text{Now, when mass is tripled } 3mg - T = 3ma \text{ and } T - \frac{B^2 l^2 v}{R} = ma$$

$$\text{On solving, } 3mg - \frac{B^2 l^2 v}{R} = 4ma \quad \dots (i)$$

$$\text{Now, } v = v_T \text{ when } a = 0 \Rightarrow v_T = \frac{3mgR}{B^2 l^2} = 30\text{ m/s}$$

Question Stem for Question Nos. 16 and 17
Question Stem

A solid cylinder of mass m and radius R is kept at rest on a plank of mass $2m$ lying on a smooth horizontal surface. Massless and inextensible string connecting cylinder to the plank is passing over a massless pulley. The friction between the cylinder and the plank is sufficient to prevent slipping. Pulley A is pulled with a constant horizontal force F . [$F = 100\text{ N}$, $m = 1\text{ kg}$]



16. Acceleration of cylinder with respect to earth is

Ans. 42.86

17. Acceleration of plank with respect to earth is

Ans. 28.57

Question Stem for Question Nos. 18 and 19

Question Stem

A light of wavelength λ is incident on a metal sheet of work function $\phi = 2\text{eV}$. The wavelength λ varies with time as $\lambda = 3000 + 40t$, where λ is in $\text{\AA}$ and t is in second. The power incident on metal sheet is constant at 100 W. This signal is switched on and off for time intervals of 2 minutes and 1 minute respectively. Each time the signal is switched on, the λ start from initial value of $3000\text{\AA}$. The metal plate is grounded and electron clouding is negligible. The efficiency of photoemission is 1% ($hc = 12400\text{ eV}\text{\AA}$).

18. The time after which photomission will stop is

Ans. 80.00

Sol. Photo emission will stop if $\lambda = \lambda_0 = \frac{hc}{\phi}$

$$3000 + 40t_0 = \frac{12400}{2}, t_0 = 80\text{ s}$$

19. If the total number of photoelectrons ejected in one hour is $N \times 10^{20}$, value of 'N' is

Ans. 37.10

Sol. Rate of photo emission at time t

$$\frac{dN}{dt} = \frac{1}{100} \times \frac{100}{19840 \times 10^{-19}} [3000 + 40t]$$

$$N = \frac{1}{19840 \times 10^{-19}} [3000t + 20t^2]_0^{80}$$

$$N = 18.55 \times 10^{19} = 1.855 \times 10^{20}$$

This is the number of photoelectron ejected in 1 blinking.

In 1 hr, total number of photoelectron ejected is
 $= 1.855 \times 10^{20} \times 20 = 3.71 \times 10^{21}$

Chemistry

PART – II

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

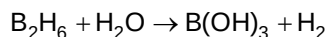
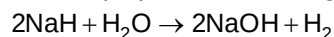
20. Which of the following is NOT correct about hydrides ?

- I. Saline hydrides are stoichiometric and metallic hydrides are non-stoichiometric.
- II. BeH_2 is monomeric whereas MgH_2 is polymeric.
- III. Hydrides of the elements of Group 13 are electron deficient.
- IV. NaH reacts with water and liberates H_2 whereas B_2H_6 does not react with water.

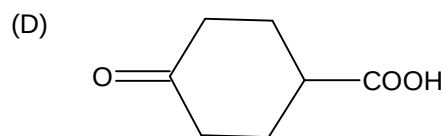
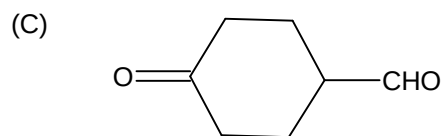
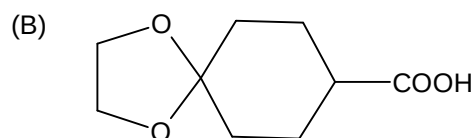
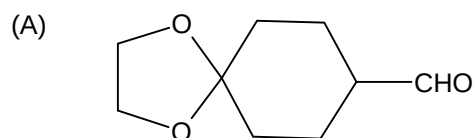
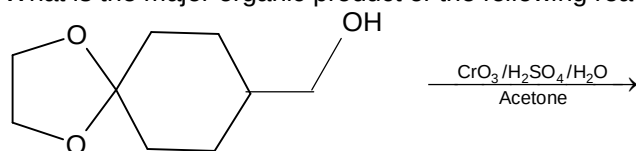
- (A) IV only
- (B) I and III
- (C) III only
- (D) II and IV

Ans. D

Sol. BeH_2 is polymeric while MgH_2 is monomeric

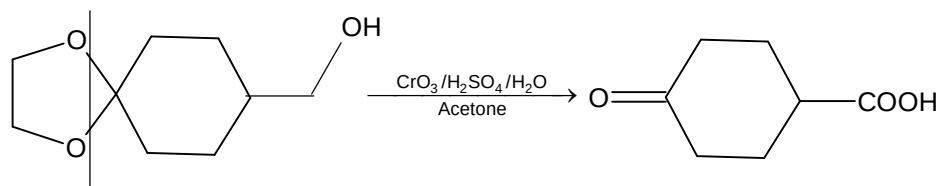


21. What is the major organic product of the following reaction?



Ans. D

Sol.



22. The solubility product of salt AB (M.W = 100) is $4 \times 10^{-10} \text{ M}^2$ at 25°C . Loss in weight of precipitate of AB by washing it with 5 litre of water is

- (A) 0.1 g
(B) 1 g
(C) 10^{-4} g
(D) 0.01 g

Ans. D

Sol. $\text{AB} \rightleftharpoons \text{A}^+ + \text{B}^-$

$$s = \sqrt{K_{sp}} = 2 \times 10^{-5} \text{ moles/lit}$$

$\therefore$ in 5 litres of water $5 \times (2 \times 10^{-5})$ moles of AB will dissolve.

$$= 10^{-4} \text{ moles}$$

$$= 10^{-4} \times 100 \text{ g}$$

$$= 0.01 \text{ g}$$

23. pH of mixture having 0.1 M NH_4OH and 0.1 M NH_4Cl is equal to $\text{pK}_w - \text{pK}_b$. If 0.1 M NH_4OH and 0.1 M $(\text{NH}_4)_2\text{SO}_4$ are taken, the pH will be

- (A) $\text{pK}_w - \text{pK}_b$
(B) $\text{pK}_w + \text{pK}_b$
(C) $\text{pK}_w - \text{pK}_b + \log 2$
(D) $\text{pK}_w - \text{pK}_b - \log 2$

Ans. D

Sol. Case-I: $\text{pOH} = \text{pK}_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_4\text{OH}]} = \text{pK}_b \quad \because [\text{NH}_4^+] = [\text{NH}_4\text{OH}] = 0.1$

$$\text{pH} = \text{pK}_w - \text{pK}_b$$

Case-II $\text{pOH} = \text{pK}_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_4\text{OH}]} \quad (\because [\text{NH}_4^+] = 0.1 \times 2)$

$$\text{pOH} = \text{pK}_b + \log 2$$

$$\text{pH} = \text{pK}_w - \log 2$$

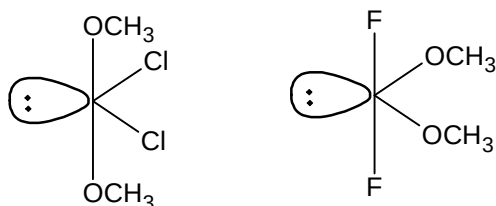
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

24. Which of the following statements is/are not correct for following compounds
 (I) $\text{SCl}_2(\text{OCH}_3)_2$ and (II) $\text{SF}_2(\text{OCH}_3)_2$
- (A) OCH_3 groups in both cases occupy the same position
- (B) Cl-atoms occupy equatorial position in case of (I) and F-atoms occupy equatorial position in case of (II)
- (C) Cl-atoms occupy axial position in case of (I) and F-atoms occupy equatorial position in case of (II)
- (D) Cl and F atoms all occupy axial position in case of both (I) and (II).

Ans. A, B, C, D

Sol.



25. Identify the incorrect statement/s
- (A) Combustion of methane in an adiabatic rigid container will cause no change in temperature of the system.
- (B) Dimerization of acetic acid causes an increase in entropy of the system
- (C) It is possible to have both adiabatic reversible and adiabatic irreversible processes between two states.
- (D) $P - V$ work is always non-zero when there is a change in volume.

Ans. A, B, C, D

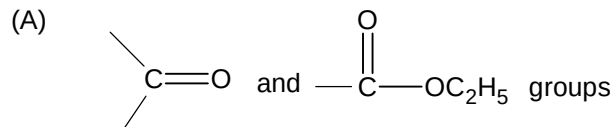
- Sol. (A) Combustion of CH_4 is exothermic so temperature will increase
 (B) Entropy decreases during dimerization.

26. According to kinetic theory of gases (more than one correct answers):
- (A) Collisions are always elastic
- (B) heavier molecules transfer more momentum to the wall of container
- (C) Only a small number of molecules have high velocity
- (D) Between collision, the molecule move in straight line with constant (average) velocities

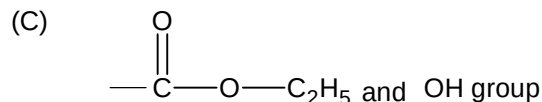
Ans. A, B, C, D

Sol. Factual

27. Ethyl acetoacetate gives the test of:



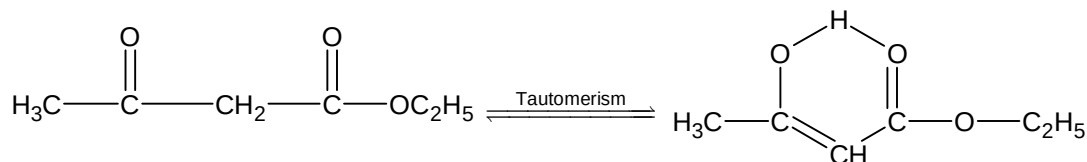
(B) $\text{CH}=\text{CH}$ and OH groups



(D) —CHO group

Ans. A, B, C

Sol.



28. Which of the following statement(s) is/are correct?

- (A) Pot. Tetracyanocuperate (I) tetrahedral complex
- (B) Sod. Tetracyanonickelate (II) is a square planar complex
- (C) Pot. Tetrachloridopatinate (II) is a tetrahedral complex
- (D) Tetraamminecopper (II) sulphate is a square planar complex

Ans. A, B, D

Sol. $[\text{Cu}(\text{CN})_4]^{3-} \rightarrow \text{tetrahedral}(\text{sp}^3)$

$[\text{Ni}(\text{CN})_4]^{2-} \rightarrow \text{square planar}(\text{dsp}^2)$

$[\text{PtCl}_4]^{2-} \rightarrow \text{square planar}(\text{dsp}^2)$

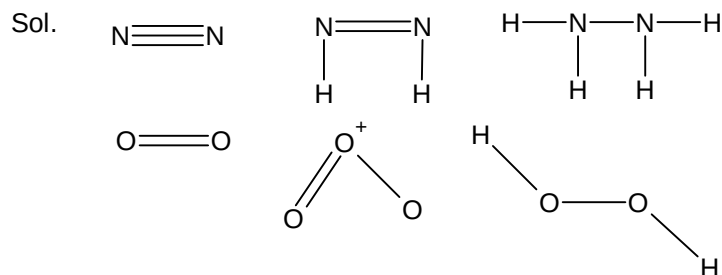
$[\text{Cu}(\text{NH}_3)_4]^{2+} \rightarrow \text{square planar}(\text{dsp}^2)$

Hence, (A), (B) and (D) are the correct answer.

29. Select CORRECT order of bond energy.

- (A) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ (halogen-halogen)
- (B) $\text{N}_2 > \text{N}_2\text{H}_2 > \text{N}_2\text{H}_4$ (N—N)
- (C) $\text{O}_2 > \text{O}_3 > \text{H}_2\text{O}_2$ (O—O)
- (D) $\text{C}_2\text{H}_2 > \text{C}_2\text{H}_4 > \text{C}_2\text{H}_6$ (C—C)

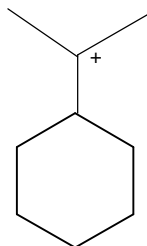
Ans. B, C, D



Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. What is the number of hyperconjugative structures of the given carbocation



Ans. 7

Sol. Number of hyperconjugative structure = number of alpha Hydrogen.

31. The van der Waals constant 'b' of a gas is $3.42 \times 10^{-2} \text{ L mol}^{-1}$. How near can be centres of the two molecules approach each other with respect to their nuclei (in Å)?

Ans. 3

Sol. $b = 3.42 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}$

$$b = 4 \times \frac{4}{3} \pi r^3 \times N_A = 3.42 \times 10^{-5}$$

$$\therefore r^3 = 3.38 \times 10^{-10} \text{ m} = 1.5 \text{ Å}$$

$$\therefore 2r = 3 \text{ Å}$$

32. In a first order reaction, the conc. of reactant decreases from 0.8 M to 0.4 M in 15 minutes. The time taken for the conc. to change from 0.1M to 0.025M is:

Ans. 30

Sol. Change of conc. from 0.8M to 0.4M reveals $t_{1/2}$ to be 15 min.

Change from 0.1M to 0.025M requires two half lives. So, the time required
 $= 2 \times 15 = 30 \text{ min.}$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34**Question Stem**

An enclosed glass bulb of volume V contains a pinch of solid NH_4Cl and $\frac{1}{100}$ moles of Ne gas. At 300 K, the internal pressure inside the glass bulb is 114 mm Hg which increases to 908 mm Hg when the bulb is heated to 600 K. Assume ideal gas behaviour.

33. What will be the partial pressure of NH_3 (g) (in mm of Hg) in the bulb at 600K

Ans. 340.00

34. What mass of NH_4Cl (in grams) is approximately present in 'Pinch'

Ans. 0.80

Sol. (Q.33-34)

Volume of bulb (V) can be calculated as per data given at 300 K

$$PV = nRT$$

$$\frac{114}{760} \times V = \frac{1}{100} \times 0.0821 \times 300$$

$$V = 1.642 \text{ lit.}$$

Now at 600 K when NH_4Cl would have decomposed

$$PV = nRT$$

$$\frac{908}{760} \times 1.64 = n \times 0.082 \times 600$$

$$n = 0.039 \text{ moles}$$

$$\text{NH}_3 + \text{HCl} = 0.039 - 0.01 = 0.029 \text{ moles}$$

$$\text{NH}_3 = \frac{0.029}{2} = 0.0145 \text{ moles}$$

$$\text{HCl} = \frac{0.029}{2} = 0.0145 \text{ moles}$$

$$P_{\text{NH}_3} = \frac{0.0145}{0.039} \times 908 \approx 340 \text{ mm}$$

Question Stem for Question Nos. 35 and 36**Question Stem**

3.6 gm of Mg is burnt in limited supply of oxygen. The residue was treated with 100 ml of H_2SO_4 (35% by mass, 1.26 g/ml density), 2.463 litre of H_2 at 760 mm at 27°C was evolved. Assume MgSO_4 is completely soluble.

35. Find the % of mass of final H_2SO_4

Ans. 22.58

Sol. $\text{Mg}(0.15 \text{ moles}) + \frac{1}{2}\text{O}_2 \longrightarrow \text{MgO}$
 $\text{Mg} + \text{H}_2\text{SO}_4 \longrightarrow \text{MgSO}_4 + \text{H}_2 \uparrow (0.1 \text{ moles})$
 $\text{MgO} + \text{H}_2\text{SO}_4 \longrightarrow \text{MgSO}_4 + \text{H}_2\text{O}$
 Moles of Mg left in first reaction = 0.1 moles
 Moles of MgO formed in first reaction = 0.05 moles
 Moles of H_2SO_4 initially taken = 0.45
 Moles of H_2SO_4 net consumed = 0.15
 H_2SO_4 left = 0.30 moles
 Water left = water present initially + formed = 81.9 + 0.9 = 82.8g
 MgSO_4 formed = 18g
 Total mass of final solution = 0.3(98) + 82.8 + 18 = 130.2
 $\% \text{ by mass of final } \text{H}_2\text{SO}_4 = \frac{0.3 \times 98}{105} \times 100 = 22.58 \%$

36. Volume of O_2 (in litre) taken initially at STP will be:

Ans. 0.56

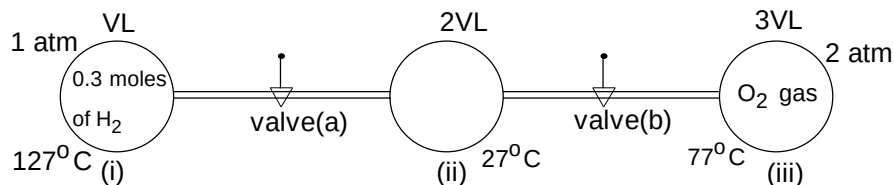
Sol. For volume of O_2 consumed at STP

$$\begin{aligned} \text{Moles taken} &= (\text{moles of MgO formed}) \times \frac{1}{2} \\ &= 0.05 \text{ moles} \times \frac{1}{2} = 0.025 \\ \text{Volume of } \text{O}_2 &= 22.4 \times 0.025 = 0.56 \text{ litre} \end{aligned}$$

Question Stem for Question Nos. 37 and 38

Question Stem

Three bulbs (I), (II), and (III) of volume V litres, 2V litres and 3V litres are connected by tubes of negligible volume fitted by stopping valve (as shown in fig). The bulb I is at 1 atm containing 0.3 moles of H_2 gas at temp 127°C . Bulb (III) contains O_2 gas at 77°C and pressure of 2 atm. Bulb (II) is vacuum and maintained at temp. 27°C .



The valve (b) is opened first and then closed after some time. Then the valve (a) is opened and closed after same time as for valve (b). The number of moles of O_2 diffused into bulb (I) is 0.1 mole

37. The partial pressure (in atm) of H_2 observed in bulb (I) after diffusion is

Ans. 0.38

Sol. Number of moles of H_2 (g) remaining in bulb (I) = 0.3 – 0.185 = 0.115

$$\therefore P \propto n \text{ (at constant T \& V)}$$

$$\therefore \frac{P_{\text{after diffusion}}}{P_{\text{initial}}} = \frac{\text{no. of moles of H}_2 \text{ after diffusion}}{\text{no. of moles initially present}}$$

$$\therefore \text{Pressure in bulb (I) after diffusion} = P_{\text{initial}} \times \frac{0.115}{0.3} = 1 \times \frac{0.115}{0.3} = 0.38 \text{ atm}$$

38. The pressure (in atm) observed in bulb (III) after diffusion is

Ans. 1.90

Sol. 37 - 38.

From bulb (I)

$$PV = nRT$$

$$V = \frac{nRT}{P} = \frac{0.3 \times 0.082 \times 400}{1}$$

$$V = 0.3 \times 0.082 \times 400 \text{ L}$$

Calculation of number of moles of H₂ diffused in bulb (II)

$$\frac{r_{\text{H}_2}}{r_{\text{O}_2}} = \frac{n_{\text{H}_2 \text{ diffused}}}{n_{\text{O}_2 \text{ diffused}}} = \frac{P_{\text{H}_2}}{\sqrt{M_{\text{H}_2} \cdot T_{\text{H}_2}}} \cdot \frac{\sqrt{M_{\text{O}_2} \cdot T_{\text{O}_2}}}{P_{\text{O}_2}} \quad \left(\because r = \frac{PA}{\sqrt{2\pi MRT}} \right)$$

$$\frac{n_{\text{H}_2 \text{ diffused}}}{n_{\text{O}_2 \text{ diffused}}} = \frac{n_{\text{H}_2}}{0.1} = \frac{1}{\sqrt{2} \times 100} \times \frac{\sqrt{32 \times 350}}{2} = \frac{\sqrt{14}}{2} = \frac{3.7}{2} = 1.85$$

$\therefore$ number of moles of H₂ diffused in bulb (II) = 0.185

For bulb (III)

$$\text{Volume} = 3V = 3 \times 0.3 \times 0.082 \times 400 \text{ L}$$

$$\text{Number of moles of O}_2 \text{ initially in bulb (III)} = \frac{PV}{RT} = \frac{2 \times 3 \times 0.3 \times 0.082 \times 400}{0.082 \times 350} = 2.05$$

$$\text{Number of moles of O}_2 \text{ remaining after diffusion} = 2.05 - 0.1 = 1.95$$

$\therefore P \propto n$ (at const. T & V)

$$\therefore \frac{P_{\text{initial}}}{P_{\text{after diffusion}}} = \frac{2.05}{1.95}$$

$$P_{\text{after diffusion}} = P_{\text{initial}} \times \frac{1.95}{2.05} = 2 \times \frac{1.95}{2.05} = 1.9 \text{ atm}$$

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

39.
$$y = \frac{(e^{\sin x} - 1)^4 + (\cos x - 1)^3 e^{\sin x} + (e^x - e^{-x})^3 + x^3 + 2x^2 + 4}{e^x}$$

Then the value of $y''(0)$ is

- (A) -4
- (B) 4
- (C) -8
- (D) 8

Ans. D

Sol. At $x = 0, y = 4$

$$e^x y = x^3 + 2x^2 + 4$$

$$\Rightarrow y''(0) = 8$$

$$y''(0)$$

Because $y''(0), (e^{\sin x} - 1)^4 + (\cos x - 1)^3 e^{\sin x} + (e^x - e^{-x})^3$ is always gives zero at $x = 0$

40. If p, q, r are prime numbers and α, β, γ are positive integers such that L.C.M. of α, β, γ is $p^3 q^2 r$ and greatest common divisor of α, β, γ is pqr then the number of possible triplets (α, β, γ) will be

- (A) 6
- (B) 12
- (C) 72
- (D) 96

Ans. C

Sol. LCM of $\alpha, \beta, \gamma = p^3 q^2 r$ and H.C.F. = pqr

$$\text{Let } \alpha = p^{m_1} q^{n_1} r, \beta = p^{m_2} q^{n_2} r \text{ and } \gamma = p^{m_3} q^{n_3} r.$$

Minimum of $(m_1, m_2, m_3) = 1$ and maximum of $(m_1, m_2, m_3) = 3$.

$$\therefore \text{Number of possibilities for } m_1, m_2, m_3 = 3 + 3 + 3 = 12$$

$$\text{Similarly, number of possibilities for } n_1, n_2, n_3 = 3 + 3 = 6$$

$$\text{And number of possibilities for } r_1, r_2, r_3 = 1$$

$$\text{Total triplets } 12 \times 6 \times 1 = 72$$

41. Each of the nine balls can be placed with equal probability in any of the three initially empty boxes. Let P_1 be the probability that there will be four balls in the first box, three in the second box and two in the third box, and P_2 be the probability that three will be in each box then the ratio (P_2 / P_1) equal to.

- (A) $\frac{4}{3}$
 (B) $\frac{2}{9}$
 (C) 8
 (D) 2

Ans. A

Sol. $P_1 = \frac{9!}{4!3!2!} \cdot \frac{1}{3^9}$ (4 must go in B_1 , 3 in B_2 and 2 in B_3)

And $P_2 = \frac{9!}{(3!)^3} \cdot \frac{1}{3^9}$ all boxes are different

$$\therefore \frac{P_2}{P_1} = \frac{4}{3}$$

42. If a, b, c are distinct positive real number such that $b(a + c) = 2ac$, then the roots of $ax^2 + 2bx + c = 0$ are

- (A) real and equal
 (B) real and distinct
 (C) imaginary
 (D) none of these

Ans. C

Sol. $b = \frac{2ac}{a+c}$ that means b is the H.M. of a and c .

$$\Rightarrow ax^2 + 2bx + c = 0$$

$$\Rightarrow D = 4(b^2 - ac)$$

Since G.M. of a and c is $\sqrt{ac}$

$$\therefore ac > b^2$$

$$\Rightarrow D < 0.$$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

43. If a set 'A' is $\{1, 0, -1, 2\}$ then which of the following is(are) correct?
- (A) A square matrix of order 3 is constructed whose elements are from set 'A' then the probability that matrix is skew matrix is $\frac{27}{4^9}$.
- (B) A square matrix of order 3 is constructed whose elements are from set 'A' then probability that matrix is symmetric is $\frac{1}{64}$.
- (C) A function $f : A \rightarrow A$ is defined then probability that function is onto is $\frac{3}{32}$.
- (D) A function $f : A \rightarrow A$ is defined and f is onto then the probability that $f(i) \neq i \forall i \in A$ is $\frac{3}{8}$.

Ans. A, B, C, D

Sol. (A) In skew symmetric matrix $a_{ij} = -a_{ji}$ and all element of diagonal are zero

$$\text{So required probability} = \frac{3 \times 3 \times 3}{4^9} = \frac{27}{4^9}$$

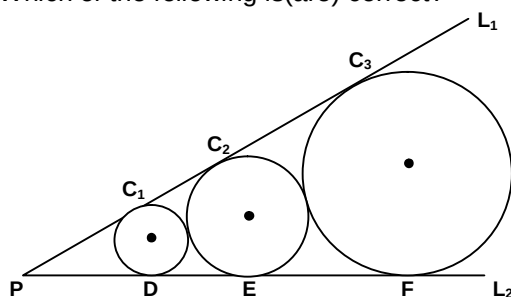
(B) For symmetric matrix $a_{ij} = a_{ji}$

$$\text{So, required probability} = \frac{(4)(4)(4)(4)(4)(4)(1.1.1)}{4^9} = \frac{1}{64}$$

$$(C) \text{ Probability } \frac{4!}{4^4} = \frac{3}{32}$$

$$(D) \text{ Probability} = \frac{4! \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right)}{4!} = \frac{9}{24} = \frac{3}{8}.$$

44. Three circles C_1, C_2, C_3 with radii r_1, r_2, r_3 ($r_1 < r_2 < r_3$) touches each other externally as shown and L_1, L_2 are two common tangents to these circles from P. Which of the following is(are) correct?



$$(A) \quad r_2 = \frac{r_1^2 + r_3^2}{r_1 + r_3}$$

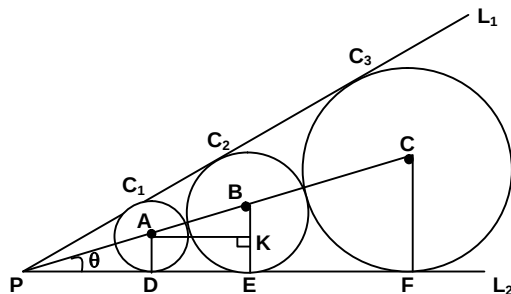
(B) $r_2^2 = r_1 r_3$

(C) $DE = 2\sqrt{r_1 r_2}$

(D) $DE = \sqrt{r_2^2 - r_1^2}$

Ans. B, C

Sol. $DE = AK = \sqrt{(r_1 + r_2)^2 - (r_2 - r_1)^2} = 2\sqrt{r_1 r_2}$


 Let $PD = x$ and $\angle APD = \theta$

$$\therefore \tan \theta = \frac{AD}{x} = \frac{BE}{x + DE} = \frac{CF}{x + DE + EF}$$

$$\frac{r_1}{x} = \frac{r_2}{x + 2\sqrt{r_1 r_2}} = \frac{r_3}{x + 2\sqrt{r_1 r_2} + 2\sqrt{r_2 r_3}}$$

$$\Rightarrow \frac{r_2 - r_1}{2\sqrt{r_1 r_2}} = \frac{r_3 - r_2}{2\sqrt{r_2 r_3}} \Rightarrow r_2^2 = r_1 r_3$$

 45. Let $g(x) = ax^2 + bx + c$, $a, b, c \in \mathbb{N}$ and $\int_0^1 g(x) dx = \frac{11}{6}$. Let $f(x)$ be a continuous and derivable

 function in (x_1, x_2) . If $f(x) \cdot f'(x) \geq x\sqrt{1 - (f(x))^4}$ and $\lim_{x \rightarrow x_1^+} (f(x))^2 = 1$ and $\lim_{x \rightarrow x_2^-} (f(x))^2 = \frac{1}{2}$, then

 minimum value of $[x_1^2 - x_2^2]$ is equal to,

 Where $[.]$ denotes greatest integer function.

(A) $2 - abc$

(B) $a + b - c$

(C) $b - c$

(D) $a - b$

Ans. A, B

Sol. $\int_0^1 g(x) dx = \frac{11}{6} \Rightarrow \frac{a}{3} + \frac{b}{2} + c = \frac{11}{6} \Rightarrow 2a + 3b + 6c = 11$

$$\therefore a = b = c = 1 \left\{ \because a, b, c \in \mathbb{N} \right.$$

$$\therefore \frac{f(x)f'(x)}{\sqrt{1 - (f(x))^4}} - x \geq 0 \Rightarrow h(x) = \sin^{-1}(f(x))^2 - x^2 \text{ is non-decreasing function}$$

$$\therefore \lim_{x \rightarrow x_1^+} h(x) \leq \lim_{x \rightarrow x_2^-} \Rightarrow \frac{\pi}{2} - x_1^2 \leq \frac{\pi}{6} - x_2^2 \Rightarrow x_1^2 - x_2^2 \geq \frac{\pi}{3} \Rightarrow [x_1^2 - x_2^2] \geq 1$$

i.e., $[x_1^2 - x_2^2] = 1$.

46. Given that $\vec{a}$ and $\vec{b}$ are two unit vectors such that angle between $\vec{a}$ and $\vec{b}$ is $\cos^{-1}\left(\frac{1}{4}\right)$. If $\vec{c}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$, such that $|\vec{c}| = 4$, $\vec{c} \times \vec{b} = 2\vec{a} \times \vec{b}$ and $\vec{c} = \lambda\vec{a} + \mu\vec{b}$ then,

- (A) $\lambda = 2$
 (B) sum of values of $\mu = -1$
 (C) product of all possible values of $\mu = -12$
 (D) number of distinct values of μ is three.

Ans. A, B, C

Sol. Given $\vec{c} \times \vec{b} = 2\vec{a} \times \vec{b} \Rightarrow (\vec{c} - 2\vec{a}) \parallel \vec{b} \Rightarrow \vec{c} = 2\vec{a} + \mu\vec{b} \dots(1)$

$$\therefore \lambda = 2$$

$$\text{Also (1)} \Rightarrow \vec{a} \cdot \vec{c} = 2 + \mu \vec{a} \cdot \vec{b} = 2 + \mu \frac{1}{4} \dots(2)$$

$$\text{Again, (1)} \Rightarrow \vec{b} \cdot \vec{c} = 2\vec{a} \cdot \vec{b} + \mu = \frac{1}{2} + \mu \dots(3)$$

$$\text{Again, (1)} \Rightarrow \vec{c} \cdot \vec{c} = 2(\vec{a} \cdot \vec{c}) + \mu(\vec{b} \cdot \vec{c}) \Rightarrow 16 = 2\left(2 + \frac{\mu}{4}\right) + \mu\left(\frac{1}{2} + \mu\right)$$

$$\Rightarrow 12 = \frac{\mu}{2} + \frac{\mu}{2} + \mu^2 \Rightarrow \mu^2 + \mu - 12 = 0$$

47. PQ and RS are normal chord to parabola $y^2 = 8x$ at P and R on the curve respectively and the points P, Q, R, S are concyclic. If A is vertex of parabola, then

- (A) centroid of ΔAPR lies on x-axis
 (B) centroid of ΔAPR lies on y-axis
 (C) PR is parallel to tangent at the vertex
 (D) PR is parallel to axis of parabola

Ans. A, C

Sol. Equation of normal chord at $P(2t_1^2, 4t_1)$ & $R(2t_2^2, 4t_2)$ are $y + t_1x - 4t_1 - 2t_1^3 = 0$ and $y + t_2x - 4t_2 - 2t_2^3 = 0$. Equation of any two degree curve passing through P, Q, R, S is $(y + t_1x - 4t_1 - 2t_1^3)(y + t_2x - 4t_2 - 2t_2^3) + \lambda(y^2 - 8x) = 0$
 $\therefore$ P, Q, R, S are concyclic
 $\therefore 1 + \lambda = t_1t_2$ & $t_1 + t_2 = 0$

$\therefore$ Centroid of ΔAPR lies on x-axis & slope of $PR = \frac{4(t_1 - t_2)}{2(t_1^2 - t_2^2)} = \frac{2}{t_1 + t_2}$

$\therefore$ PR is parallel to y-axis

48. If $f: \mathbb{R} \rightarrow [-1, 2]$, $f(x) = \frac{x^2 + 9bx + 17}{ax^3 + x^2 + bx + 33}$ is onto function and $f'(d) = f'(e) = 0$. Then

(A) eccentricity of curve $\frac{x^2}{(d+e)^2} + \frac{y^2}{b+1} = 1$ can be $\frac{1}{2}$

(B) eccentricity of curve $\frac{x^2}{(d+e)^2} + \frac{y^2}{b+1} = 1$ can be $\frac{\sqrt{5}}{2}$

(C) eccentricity of curve $\frac{x^2}{(d+e)^2} + \frac{y^2}{(a+1)} = 1$ can be $\frac{\sqrt{3}}{2}$

(D) eccentricity of curve $\frac{x^2}{(d+e)^2} + \frac{y^2}{a+1} = 1$ can be $\frac{\sqrt{7}}{2}$

Ans. A, B, C

Sol. $\therefore$ Domain of function is real number so $a = 0$

$$\text{Now } y = \frac{x^2 + 9bx + 17}{x^2 + bx + 33} \Rightarrow (y-1)x^2 + (by-9b)x + (33y-17) = 0$$

$$\therefore x \in \mathbb{R} \text{ so } D \geq 0 \Rightarrow (by-9b)^2 - 4(y-1)(33y-17) \geq 0$$

$$\therefore b = 2 \text{ and } -2$$

$$\therefore \text{For } b = 2 \Rightarrow d = -5, e = 7$$

$$\text{For } b = -2 \Rightarrow d = -7, e = 5$$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. A 3×3 determinant has entries either 1 or -1. Let S be the set of all determinants such that the

product of elements of any row or column is -1. For example $\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix}$ is an element of S and

number of elements in S is m.

Let $P = \begin{bmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{bmatrix}$ and trace of the matrix $\text{adj}(\text{adj } P)$ is n then the value of $\frac{m}{n}$ is

Ans. 8

Sol. $\therefore$ Elements of determinant can be 1 or -1

$\therefore$ First two places of first row can be filled in $2 \times 2 = 4$ ways and last place can be filled in only one way to product be 1

$\therefore$ First row can be made in 4 ways

Similarly second row can be made in 4 ways and third row can be made in 1 ways only

$$\therefore m = 16$$

$$\therefore |P| = \begin{vmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix} = 1$$

$$\therefore \text{adj}(\text{adj}P) = |P|^{n-2} P = P \Rightarrow \text{tr}(\text{adj}(\text{adj}P)) = \text{tr}(P) = 2 = n.$$

$$\therefore \frac{m}{n} = \frac{16}{2} = 8$$

50. A normal coin is tossed six times. If $p = \frac{a}{b}$ where a, b are relatively prime positive integers, denotes the chance that there will be exactly one sequence of 3 consecutive heads in six tosses, then the value of $\left(\frac{a+b+1}{4}\right)$ is

Ans. 5

Sol. H H H T X X
T H H H T X
X T H H H T
X X T H H H

$$\therefore P(E) = \frac{1}{16} + \frac{1}{32} + \frac{1}{32} + \frac{1}{16} = \frac{6}{32} = \frac{3}{16}$$

$$\frac{3+16+1}{4} = 5$$

51. If a, b, c, d, e are positive real numbers such that $a + b + c + d + e = 15$ and $ab^2c^3d^4e^5 = (120)^3 \cdot 50$, then the value of $a^2 + b^2 + c^2 + d^2 + e^2$ is

Ans. 55

Sol. A.M. =
$$\frac{a + \left(\frac{b}{2} + \frac{b}{2}\right) + \left(\frac{c}{3} + \frac{c}{3} + \frac{c}{3}\right) + \left(\frac{d}{4} + \frac{d}{4} + \frac{d}{4} + \frac{d}{4}\right) + \left(\frac{e}{5} + \frac{e}{5} + \frac{e}{5} + \frac{e}{5} + \frac{e}{5}\right)}{15} = 1$$

$$\text{And G.M.} = \left(a \cdot \frac{b}{2} \cdot \frac{b}{2} \cdot \frac{c}{3} \cdot \frac{c}{3} \cdot \frac{c}{3} \cdot \frac{d}{4} \cdot \frac{d}{4} \cdot \frac{d}{4} \cdot \frac{d}{4} \cdot \frac{e}{5} \cdot \frac{e}{5} \cdot \frac{e}{5} \cdot \frac{e}{5} \cdot \frac{e}{5}\right)^{\frac{1}{15}} = 1$$

$$\therefore a = \frac{b}{2} = \frac{c}{3} = \frac{d}{4} = \frac{e}{5}$$

$$\therefore a = 1, b = 2, c = 3, d = 4, e = 5$$

$$\therefore a^2 + b^2 + c^2 + d^2 + e^2 = 55$$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53
Question Stem

A ten-sided regular polygon is inscribed in a circle $|z| = 1$. $A_1 A_2 \dots A_{10}$ to its vertices, G_1 is centroid of $\Delta A_1 A_4 A_8$, G_2 be centroid of $\Delta A_2 A_6 A_9$ and G_3 be centroid of $\Delta A_3 A_5 A_7$. P is centroid of $\Delta G_1 G_2 G_3$. If O is the centre of circle.

52. Angle $\angle POA_1$ is equal to

Ans. 2.51

Sol. Consider $x^{10} = 1$
Hence $x = e^{ix2\pi/10}$ ($r = 1, 2, \dots, 10$)

$$A_1 = e^{ix2\pi/10} = \alpha_1$$

$$A_2 = e^{ix4\pi/10} = \alpha_2$$

$$A_3 = e^{ix6\pi/10} = \alpha_3$$

$$A_{10} = 1 = \alpha_{10}$$

Here centroid of $\Delta A_1 A_4 A_8$ is

$$G_1 = \frac{\alpha_1 + \alpha_4 + \alpha_8}{3}$$

$$G_2 = \frac{\alpha_2 + \alpha_6 + \alpha_9}{3}$$

$$G_3 = \frac{\alpha_3 + \alpha_5 + \alpha_7}{3}$$

$$\text{Hence centroid of } \Delta G_1 G_2 G_3 \text{ is } P = \frac{\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_9}{9} = -\frac{1}{9}$$

$$\text{Hence } \angle POA_1 = \frac{4\pi}{5}$$

53. $100 \times OP$ is equal to

Ans. 11.11

Sol. Consider $x^{10} = 1$
Hence $x = e^{ix2\pi/10}$ ($r = 1, 2, \dots, 10$)

$$A_1 = e^{ix2\pi/10} = \alpha_1$$

$$A_2 = e^{ix4\pi/10} = \alpha_2$$

$$A_3 = e^{ix6\pi/10} = \alpha_3$$

$$A_{10} = 1 = \alpha_{10}$$

Here centroid of $\Delta A_1 A_4 A_8$ is

$$G_1 = \frac{\alpha_1 + \alpha_4 + \alpha_8}{3}$$

$$G_2 = \frac{\alpha_2 + \alpha_6 + \alpha_9}{3}$$

$$G_3 = \frac{\alpha_3 + \alpha_5 + \alpha_7}{3}$$

$$\text{Hence centroid of } \Delta G_1 G_2 G_3 \text{ is } P = \frac{\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_9}{9} = -\frac{1}{9}$$

$$OP = \frac{1}{9}$$

Question Stem for Question Nos. 54 and 55

Question Stem

Define a function $\theta : \mathbb{N} \rightarrow \mathbb{N}$ as follows, $\theta(1) = 1$, $\theta(r^n) = r^{n-1}(r-1)$, if r is prime and $n \in \mathbb{N}$, $\theta(\ell \times n) = \theta(\ell)\theta(n)$, if H.C.F(ℓ, n) = 1.

54. $\theta(1672) \div 125$ is equal to

Ans. 5.76

Sol. $1672 = 209 \times 8$, whose HCF is 1

$$\therefore \theta(1672) = \theta(209)\theta(8) = \theta(209)\theta(2^3) = \theta(11 \times 19)\theta(2^3)$$

55. If $\theta(61^5) = \lambda$, then $\frac{\lambda}{1000}$ is equal to

Ans. 223.26

Sol. $\theta(61^n) = 61^{n-1} \times 60 = 61^2 \times 60$

Question Stem for Question Nos. 56 and 57

Question Stem

$f, g : \mathbb{R} \rightarrow \mathbb{R}$ are functions such that $f(x) = 3x - \sin\left(\frac{\pi x}{2}\right)$, $g(x) = x^3 + 2x - \sin\left(\frac{\pi x}{2}\right)$

56. The value of $\frac{d}{dx} f^{-1}(g^{-1}(x))_{x=12}$ is equal to $\frac{2}{3k}$ then k is equal to

Ans. 31.14

Sol. $(f^{-1}g^{-1}(x)) = (gf(x))^{-1}$

$$\frac{d}{dx} (f^{-1}g^{-1}(x)) \Big|_{x=12} = \frac{1}{\frac{d}{dx} (gof(x)) \Big|_{x=1}}$$

57. The area bounded by $y = f^{-1}(x)$, $y = g^{-1}(x)$ is

Ans. 0.50

Sol. Area will be same as bounded by $y = f(x)$, $y = g(x)$

$$\Rightarrow \int_{-1}^1 |x^3 - x| dx = \frac{1}{2}$$